ModelIt!

Student Activity Pack

Sponge vs. Pavement

Where Does the Water Go When It Rains?

Grade NE | NGSS MS-ESS2-4, MS-ESS3-3 | Cause and Effect

STUDENT NAME _____ DATE _____

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where does rain go after it hits the ground?

2 Why do you think parking lots get big puddles but forests usually do not?

3 Draw what you think happens to rain when it hits soil vs. when it hits pavement.

4 What does the word 'absorb' mean to you?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Ground Type | Rainfall Amount | Water Absorbed | Runoff

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Rain on Soil

Soil surface, heavy rain

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Rain on Pavement

Pavement surface, heavy rain

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Light vs. Heavy Rain

Soil surface, changing rainfall

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Slope Angle — How steep the ground is — steeper slopes mean water flows faster before it can soak in

Vegetation Cover — Plants and their roots help water soak into the ground and slow down runoff

Soil Saturation — How much water the soil already holds — saturated soil cannot absorb more

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

Why do some places flood when it rains and other places do not? Use evidence from your simulation to support your answers.

1. **Why does a parking lot flood more easily than a field of grass?**

2. **What happens underground when soil absorbs water?**

3. **How do cities make flooding worse by building roads and buildings?**

4. **What are some ways to add more permeable surfaces to a city?**

5. **How did the model help you predict where water goes during a storm?**

8 STEM Challenge: Schoolyard Watershed Map

ModelIt!

YOUR ENGINEERING MISSION

Map every surface type on your school grounds and predict where water collects during a rainstorm.

THE SCENARIO

The school principal wants to reduce puddles and flooding on the playground. Your team has been asked to study the school grounds, identify problem areas, and recommend solutions based on your model data.

Engineering Design Thinking

1. HOW WILL YOU MAP THE DIFFERENT SURFACE TYPES ON YOUR SCHOOL GROUNDS?
2. HOW WILL YOU MEASURE OR ESTIMATE THE AREA OF EACH SURFACE TYPE?
3. WHERE DO YOU PREDICT THE WATER WILL FLOW BASED ON YOUR MODEL?
4. WHAT CHANGES WOULD YOU RECOMMEND TO REDUCE RUNOFF?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Hydrologists study how water moves through the environment — where it flows, where it soaks in, and where it causes problems. They help cities design better drainage and prevent floods. They earn $65,000–$100,000/year.